Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / bending
Case	rca_ba7383902c6c4806b0ec1ed47141bc48
Date	2026-11-01 18:09:46
Status	AI proposal, pending expert review

1. Problem

Defect	Severity	Confidence	Detail
scratch	minor	95%	46.1 mm long
dent	major	88%	~14 mm
deformation	major	99%	bend 93.1 deg (nom 90 deg ±0.5)

2. Root cause

Cause: Tool Wear / Die Damage

Reason: The presence of a 46.1 mm scratch and a 14 mm dent directly correlates with surface defects caused by damaged or worn die surfaces as described in [587aa665-c6f1-48df-8783-09d65ff4f82a].

Effect: Deformation, Surface Defects

Confidence: 76%

Supporting evidence

- **587aa665-c6f1-48df-8783-09d65ff4f82a**: Tool Wear / Die Damage Root Cause Analysis Defect Types: scratch, dent, edge burr, dimensional shift Process: bending, stamping, forming Root Cause: Progressive wear of punch/die tooling causes dimensional drift, surface scratches, edge burrs and dents on formed parts. Effect: Surface defects (scratches from worn tooling edges), dents from damaged die surfaces, edge burrs from worn cutting edges,

3. Solution

Corrective actions (fix this occurrence)

1. Inspect tooling under magnification for wear, chipping, or damage
2. Regrind worn punch/die to restore geometry within tolerance

Preventive actions (stop it recurring)

1. Establish a periodic tooling inspection schedule
2. Monitor surface finish quality during production runs

Precautions

- Ensure die surfaces are free of debris before operation

How to verify the fix

Visual inspection of subsequent parts for absence of scratches and dents

4. Other causes considered

Rank	Cause	Confidence	Reason

2	Improper Bending Force	74%	The detected bend angle of 93.1 degrees (exceeding the 90.5 degree limit) indicates an issue with bending force settings or over-bending, which is a known cause for deformation in sheet metal bending [586f086d-0deb-40c2-a71d-f840aaa55a60].
3	Springback	53%	The deviation in the bend angle (93.1 degrees vs 90 degrees) is a classic indicator of elastic recovery (springback) occurring after the bending process, as outlined in [6872e1ff-9d0f-41f8-9cd9-c783d66c1b46].

5. Traceability

AI model: gemini:gemini-3.1-flash-lite. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.